

JC4003 Natural Language Processing

Lecture 2: Basic Language Concepts and the Diversity

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Outline

Pronunciation Features

Writing System Features

Grammatical Features

Pronunciation Features

Classification of Speech Sounds

- **Vowels** are speech sounds produced by the vibration of the vocal cords with an open vocal tract, allowing the air to flow freely without any significant obstruction.

E.g. [a], [e], [i], [o], [u]

- **Consonants** are speech sounds produced by obstructing the airflow at one or more points in the vocal tract.

E.g. [p], [t], [k], [s], [m]

Vowels

Vowels are speech sounds produced by the vibration of the vocal cords with an open vocal tract, allowing the air to flow freely without any significant obstruction.

E.g. [a], [e], [i], [o], [u]

Languages vary greatly in the number of vowel sounds they possess. Some languages have an extensive range of vowels, while others have very few.

Danish is known for its rich vowel system 12~27 vowels (12 short, 13 long, plus 2 central).

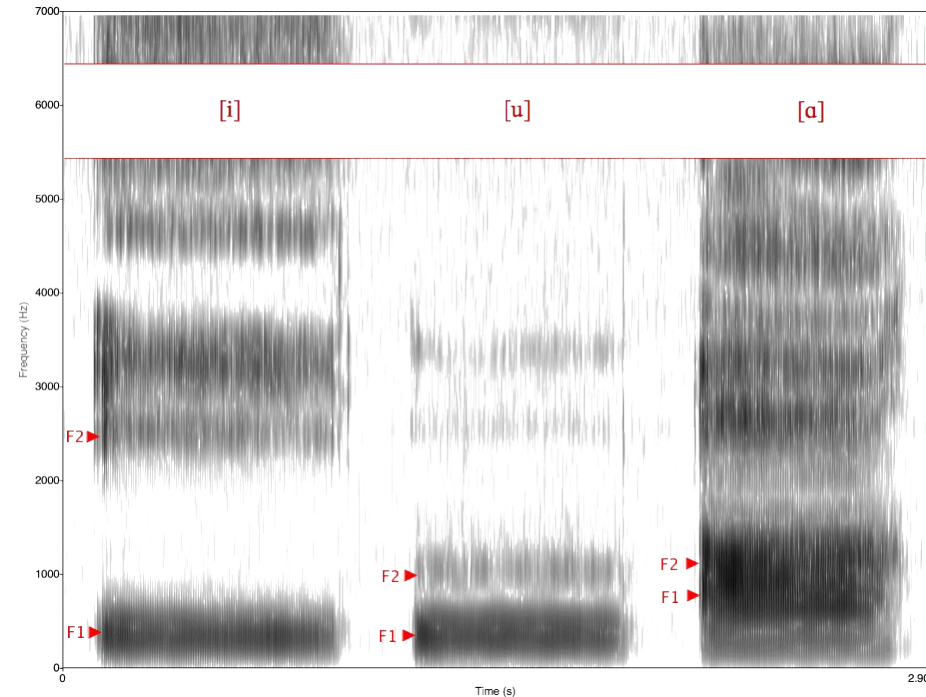
In contrast, **Ubykh**, a now-extinct language from the North Caucasus, had only 2~3 vowels phonemes, [ɐ], [ɜ], [ɨ], despite having a highly complex consonant system.

Acoustic Characteristics of Vowels

Different vowels are characterised by the relative values of **formants**, which are the acoustic resonances of the vocal tract and appear as dark bands on a spectrogram.

The vocal tract acts as a resonating chamber, and the position of the jaw, lips, and tongue alters its dimensions, thereby affecting the formant frequencies produced.

The acoustic properties of vowels can be visualised using a **spectrogram**, which displays the energy present at each frequency and its variation over time.

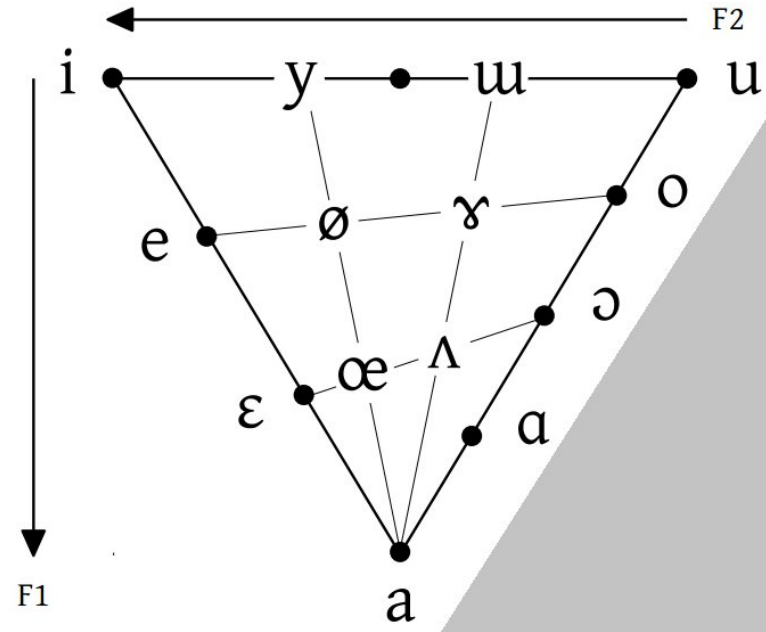


Acoustic Characteristics of Vowels

Idealized Vowel Space represents the acoustic characteristics of different vowels, particularly the distribution of formant frequencies. In the diagram, **F1** and **F2** denote the first and second **formant** frequencies (logarithmic), which are key parameters in describing vowel quality.

The grey range is where F2 would be less than F1, which by definition is impossible.

Some vowels are more complex in their composition and cannot be represented on the diagram.



Idealized Vowel Space

Vowel Length

Vowel length refers to the duration of time a vowel sound is held during pronunciation, distinguishing between short and long vowels.

English examples:

Short: bit [ɪ] vs. Long: beet [i:]

Danish examples:

	short	long	short	long	short	long	short	long
Close	i	i:	y	y:			u	u:
Close-mid	e	e:	ø	ø:	ə		o	o:
Open-mid	ɛ	ɛ:	œ	œ:	ɐ		ɔ	ɔ:
Open	a	a:			ɑ	ɑ:	ɒ	ɒ:

Diphthongs and Triphthongs

Diphthongs and triphthongs are **complex vowel sounds** found in many languages.

A **diphthong** consists of **two vowel** sounds or articulatory positions **within the same syllable**, where the tongue glides smoothly from the first vowel to the second. A **triphthong** involves **three vowel sounds**.

Diphthongs examples:

high [haɪ], cow [kaʊ], 爱 [aɪʅ], 家 [tɕjaʅ]

Triphthongs examples:

fire [faɪə], layer [leɪə], 跳 [tʰjaʊʅ], 九 [tɕjoʊʅ]

Consonants

Consonants are speech sounds produced by obstructing the airflow at one or more points in the vocal tract.

E.g. [b], [p], [m], [ŋ], [ts], [ds]

Ubykh had an astonishing **84 consonant phonemes**, including a wide variety of glottal, pharyngeal, plosive, fricative, and affricate sounds.

Rotokas, spoken in the Solomon Islands, is often cited as having the fewest consonant phonemes of any language. It is generally believed to have only **6~7 consonant phonemes**: [p], [t], [k], [b], [d], [g], with [r] considered to overlap with other phonemes.

Voiceless vs. Voiced

Voiceless sounds are produced without vibration of the vocal cords, such as [p] and [s], while voiced sounds involve vocal cord vibration, such as [b] and [z].

Voiceless

Voiced

[p]



[b]

[s]



[z]

[θ]



[ð]

Aspirated Sounds vs. Unaspirated Sounds

Aspirated sounds are produced with a noticeable puff of air.

Aspirated

top [t^hɒp]

kit [k^hɪt]

八 [p^haɪ]

vs.

Unaspirated

stop [stɒp]

skate [skeɪt]

葩 [p^haɪ]

Trilled Consonants

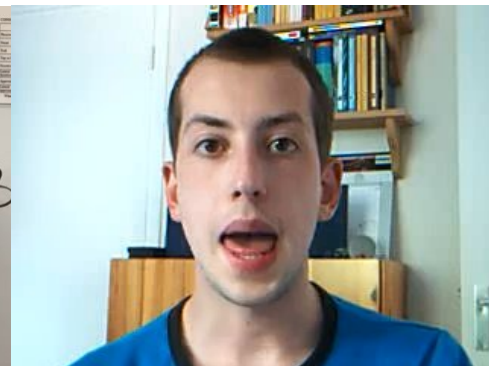
Trilled consonants, also known as trills, are produced by vibrating a part of the vocal tract rapidly against another. There are four trilled consonants.

Bilabial Trill: /b/

Alveolar Trill: /r/

Uvular Trill: /ʀ/

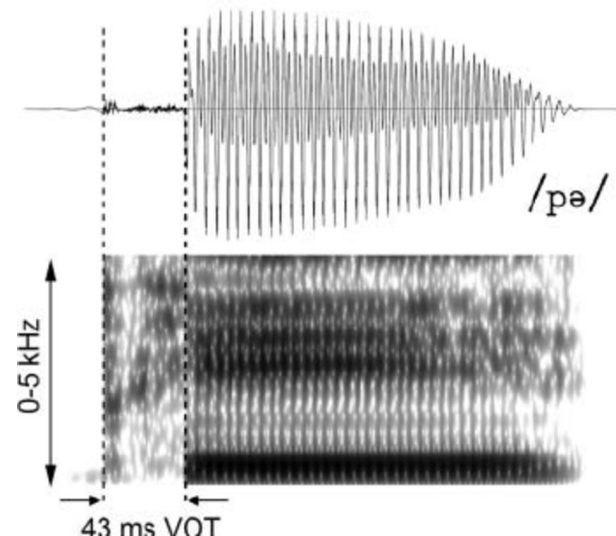
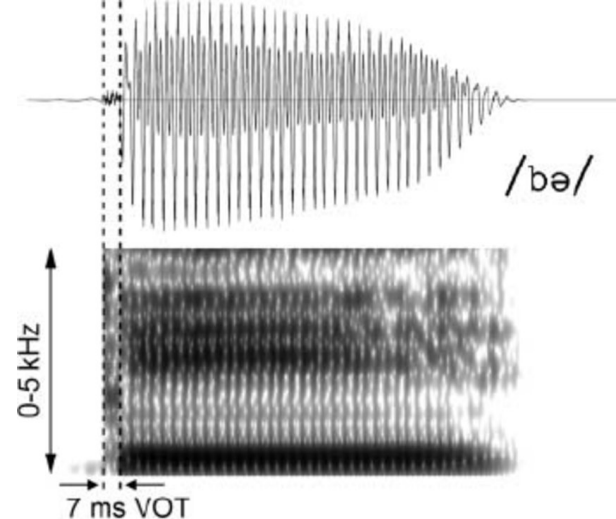
Linguolabial Trill: /ɾ/



Voice Onset Time (VOT)

Voice Onset Time (VOT) refers to the length of time that passes between the release of a consonant and the onset of vocal cord vibration, which is crucial in distinguishing between voiceless and voiced consonants in many languages.

- **Pre-voicing:** The vowel starts before the consonant is released, resulting in a negative VOT value (e.g. -10ms).
- **Short-lag:** The vowel starts shortly after the release of the consonant (e.g. VOT = 10ms).
- **Long-lag:** Here, there is a longer delay before the vowel after the release of the consonant (e.g. VOT = 50ms).



Voice Onset Time (VOT)

The differences in VOT (Voice Onset Time) can be quite significant across different languages and accents. For example, in English, the VOT for voiceless stops like [p], [t] is typically longer:

British English:	[p] = 42ms	[b] = 15ms	[t] = 64ms	[d] = 21ms
American English:	[p] = 58ms	[b] = 0/-100ms	[t] = 70ms	[d] = 0/-100ms
Chinese English:	[p] = 80ms	[b] = 15ms	[t] = 80ms	[d] = 21ms
Indian English:	[p] = 30ms	[b] = 10ms	[t] = 30ms	[d] = 10ms

NB1: These data are synthesised from various papers, and due to potential bias in the subjects studied, the data may be inaccurate.

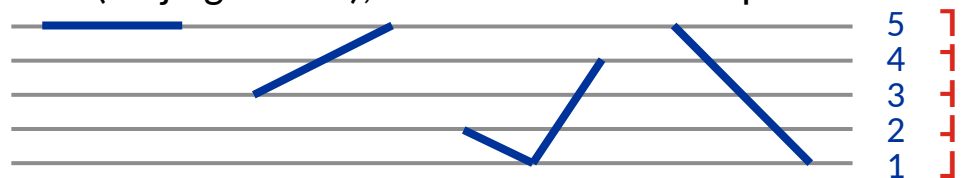
NB2: American English exhibits a phenomenon of devoicing of voiceless consonants; the table shows statistics for voiceless consonants without devoicing.

Tone

Tone is the use of pitch in language to distinguish lexical or grammatical meaning, that is, to distinguish or to inflect words. Languages that have this feature are called **tonal languages**.

The IPA incorporates Yuen Ren Chao (赵元任)'s tone notation system, using five levels to represent pitch variations in tones.

E.g., in mandarin (Beijing dialect), the four tones are represented like:



妈

麻

马

骂

ㄇㄚˊ = 1

ㄇㄚˊˊ = 11

ㄇㄚˋ = 111

ㄇㄚˋˊ = 111

IPA

55

35

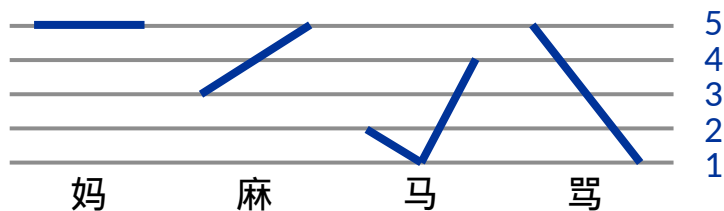
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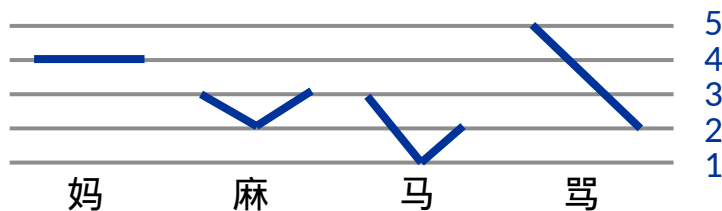
Chao's notation

Tone

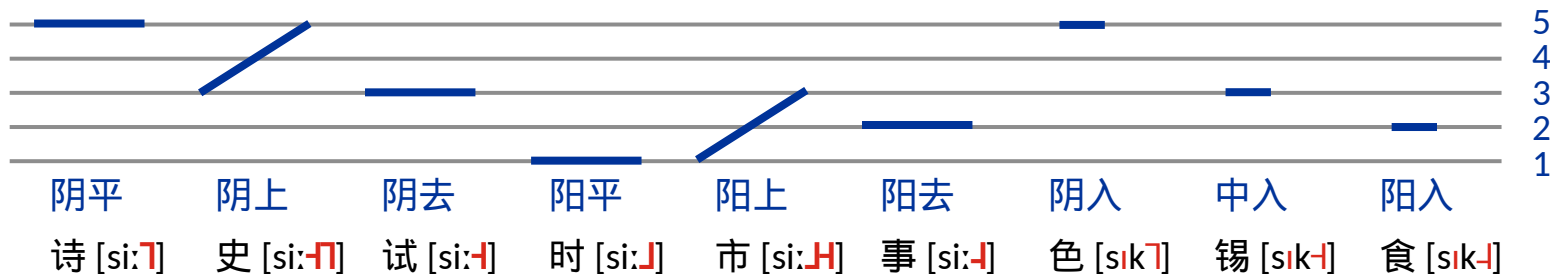
Mandarin (Beijing dialect)



Mandarin (Taipei dialect)



Cantonese (Guangzhou)



Phonological Processes

Assimilation: This occurs when a sound becomes similar or identical to a neighbouring sound.

E.g., "input" : ['ɪnpʊt] → ['ɪmpʊt]

Elision: This process involves the omission of a sound in certain contexts.

E.g. "camera" : ['kæməɹə] → ['kæmrə]

E.g. 西红柿炒鸡蛋 → [胸]柿炒鸡蛋

Epenthesis: This involves the insertion of an additional sound into a word.

E.g. "something" : ['sʌmθɪŋ] → ['sʌmpθɪŋ]

E.g. 天气预报 → 天[去]预报

Writing System Features

Writing Systems

Writing Systems are the methods of visually representing language through symbols.

Classification by basic linguistic unit:

- Syllabaries
- Alphabets
- Logographic systems
- Mixed Writing Systems

Alphabetic Writing Systems

In alphabetic systems, symbols or letters represent individual phonemes (sounds). E.g.:

- **Latin** alphabet, which is used in English and many other languages
- **Greek** alphabet, known for its use in the Greek language
- **Cyrillic** alphabet, used in Russian and several other Slavic languages.

Syllabic Writing Systems

Syllabic systems use symbols to represent syllables. Each symbol typically corresponds to a whole syllable rather than an individual sound. E,g.:

- **Kana** scripts of Japanese including Hiragana and Katakana
- **Devanagari** used in Hindi and other Indian languages

Logographic Writing Systems

Logographic systems use symbols to represent words or morphemes (the smallest units of meaning).

- Chinese
- Kanji in Japanese

Mixed Writing Systems

Some writing systems incorporate elements from more than one of these categories. E.g:

- Japanese combines logographic characters (**kanji**) and syllabaries (**kana**).

日本語には複数のカテゴリーの要素が組み込まれています

- Egyptian Hieroglyphs combine logograms, syllabaries, and determinatives

Alphabet Symbols																																																																																																																																	
L  Lion	M  Owl	N  Ripple of water	A  Vulture	B  Foot	C  Basket with handle	D  Hand	E  Reed	F  Horned Viper	G  Stand	H  Twisted Flax	I  Reed	J  Snake	K  Basket with handle	Z  Door Bolt																																																																																																																			
O  Lasso	P  Stool	Q  Hill Side	R  Mouth	S  Folded cloth	T  Loaf	U  Quail chick	V  Quail chick	W  Quail chick	X  Two reeds	Y  Two reeds																																																																																																																							
Determinative Symbols																																																																																																																																	
Male 													Female 													Father 													Mother 													Plural 																																																																													
Word Examples																																																																																																																																	
Greetings 													I / Me 													You 													Love 													Happy 													Kiss 													year 													Month 													Day 													Night 												

Grammatical Features

Word Order

Word order refers to the sequence in which words are arranged to form sentences in a language. The primary word orders are based on SVO (Subject-Verb-Object) in a sentence.

- **SVO**: very common e.g. English, Chinese
- **SOV**: very common e.g. Japanese, Korean, languages of India, modern spoken Arabic
- **VSO**: Classical Arabic, Welsh
- **VOS**: Malagasy, Fijian
- **OVS**: Hixkaryana (a Carib language)
- **OSV**: very uncommon e.g. Tobati (an Austronesian language), Warao

Word Order (cont.)

- Flexible Word Orders – Finnish has relatively free word orders, meaning that the sequence of words in a sentence can vary without changing the fundamental meaning.
- E.g. 'I love you' in Finnish can be:

Minä sinua rakastan.

Rakastan minä sinua.

Rakastan sinua minä.

Sinua rakastan minä.

Sinua minä rakastan.

Word Order (cont.)

In many languages, there are also specific grammatical constructions, such as inversion, that allow for changes in the standard word order.

- E.g. in English:
 - **VSO**: Are you coming?
 - **OSV**: This problem, we must solve.
- E.g. in Chinese:
 - **SOV**: 我把作业做了
 - **OSV**: 作业被我做了

Tense & Aspect

Tense refers to the grammatical feature that indicates the time at which an action occurs, typically categorized into **past**, **present**, and **future**.

Aspect describes the nature of an action as it relates to time, focusing on whether the action is **ongoing**, **completed**, **repeated**, or **habitual**.

While tense locates an action in time, aspect provides information about the flow or completeness of the action within that time frame, highlighting different characteristics of the action's duration or frequency.

NB: In **English** language teaching, tense and aspect are often considered as a **combined** concept because English uniquely intertwines these elements, such that verb forms typically convey both the timing of an action (tense) and its temporal characteristics (aspect) simultaneously.

NB: **Chinese** does not have grammatical markers of tense, but it has a rich system of aspects.

Tense & Aspect

Russian has 3 tenses and 2 aspects, and they are disentangled, e.g. "I (male), read, book"

	Imperfective	Perfective
Present	Я читаю книгу. <i>I am reading a book.</i>	N/A 'perfective' implies completion
Past	Я читал книгу. <i>I was reading a book.</i>	Я прочитал книгу. <i>I read (past) the book (and finished it).</i>
Future	Я буду читать книгу. <i>I will be reading the book.</i>	Я прочитаю книгу. <i>I will read (and finish) the book.</i>

Perfective aspect indicates the completed actions.

Tense & Aspect

In Chinese, aspect is primarily indicated through the use of particles, e.g. "I (male), read, book"

Aspect	Marker	Usage	Chinese
Perfective	了	completed action	我读了书
Progressive	在	in progress	我在读书
Experiential	过	been experienced or done before	我读过书
Durative	着	continuous or ongoing	我读着书
Inchoative	起/起来	beginning of an action or state	我读起书(来)

Mood

Mood is a grammatical feature that **shows the speaker's attitude**. It can express whether the speaker sees the action as a **fact, command, possibility, necessity, desire**, etc.

Moods are usually indicated by verb forms and can greatly change the meaning of a sentence.

Indicative Mood: Used for stating facts or asking questions.

"She sings." (a fact) or "Does she sing?" (a question)

Imperative Mood: Used for giving commands or requests.

"Close the window." (a command)

Subjunctive Mood: Used for expressing wishes, doubts, or hypothetical situations.

"If I were taller, I would play basketball." (a wish or hypothetical)

Mood (cont.)

Conditional Mood: Used to talk about things that would happen if a certain condition is met.

"I would go if I could." (a condition)

Hortative Mood: Used for encouraging a group (with the speaker) to perform an action

"가자!" (Let's go!, a suggestion with encouraging)

Hypothetical Mood: Used to discuss situations that are not real but imagined or speculative.

"雨が降ったら、行きません。" (If it rains, I won't go.)

Other moods refer to https://en.wikipedia.org/wiki/Category:Grammatical_moods

Voice (Active Voice / Passive Voice)

In various languages, transforming an active sentence into a passive one involves altering the verb form and reorganising the sentence structure to highlight the recipient of the action rather than the doer. Each language has its unique morphological or syntactic rules for constructing passive voice, reflecting its grammatical structure.

Languages that express the passive voice through verb conjugation or inflection include, e.g., [English](#), [German](#), [French](#), [Spanish](#), [Italian](#), [Russian](#), [Greek \(Modern\)](#), [Japanese](#).

[Nynorsk](#) and [Icelandic](#) have a similar system, but the usage of the passive is more restricted.

In [Chinese](#), passive meaning is conveyed using the particle "被" before the verb.

Person

In grammar, person refers to the distinctions between different participants in a conversation or discourse, typically categorised as **first person**, **second person**, and **third person**. These categories help identify the speaker, the addressee, and others who are neither the speaker nor the addressee.

- The **first person** refers to the speaker or speakers in a conversation. E.g. **I, we**
- The **second person** refers to the person or people being addressed. E.g. **you**
- The **third person** refers to individuals or entities being talked about. E.g. **he, she, they**

In some languages, additional grammatical persons are recognized. For example, certain Algonquian languages include a **fourth person**.

Gender

Grammatical gender refers to a system of **noun classification**. Many languages use gender to categorise nouns, pronouns, and sometimes adjectives and verbs.

NB: the concept of gender in language is not always aligned with biological sex.

- In English, pronouns are linked to gender, with "**he**" typically used for males and "**she**" for females.
- In Chinese, "**他**" also used for males and "**她**" for females.

Some languages can have multiple genders, e.g. Bantu languages can have 10-20 genders.

Gender (cont.)

Gender often causes related words, such as adjectives, pronouns, and verbs, to change their forms to agree with the gender of the noun they modify or refer to. E.g.:

In **French**, the past participle of verbs must agree in gender and number with the preceding direct object when used in compound tenses like the perfect tense. For example:

Il est **allé** au parc. (**He** went to the park.)

Elle est **allée** au parc. (**She** went to the park.)

Gender (cont.)

In **Swahili**, a Bantu language, the noun gender system influences the form of adjectives, verbs, pronouns, and other related words to ensure agreement with the noun. E.g.:

person (class 1)	M <u>tu</u> m <u>zuri</u> <u>m</u> moja yule a <u>me</u> anguka.	<u>one</u> good person fells down.
people (class 2)	W <u>atu</u> w <u>azuri</u> <u>w</u> awili wale w <u>a</u> meanguka.	<u>two</u> good people fell down.
basket (class 4)	K <u>ikapu</u> k <u>izuri</u> <u>k</u> imoja kile k <u>i</u> meanguka.	<u>one</u> good basket fells down.
baskets (class 5)	V <u>ikapu</u> v <u>izuri</u> <u>vi</u> wili vile v <u>i</u> meanguka.	<u>two</u> good baskets fell down.

Case

In linguistics, **case** refers to the grammatical category that marks the relationship between a noun (or pronoun) and other elements in a sentence.

It typically manifests through **changes in the form of the noun or pronoun**, often by inflection or the use of specific prepositions or particles.

In **English**, there are primarily three grammatical cases, though they are marked in **pronouns** rather than in nouns or adjectives.

- **Nominative Case (Subjective Case):** I, you, he, she, it, we, they
- **Accusative Case (Objective Case):** me, you, him, her, it, us, them
- **Genitive Case (Possessive Case):** my/mine, your/yours, his, her/hers, its, our/ours, their/theirs; also, the 's e.g. the teacher's lounge

Case (cont.)

Finnish has 15 cases. They cover a wide range of syntactic and semantic roles, including but not limited to:

Nominative: Used for the subject of a sentence.

Accusative: Marks the direct object.

Genitive: Indicates possession.

Abessive: Indicates absence.

Inessive: Indicates location within something.

Elative: Indicates movement out of something.

Illative: Indicates movement into something.

Partitive: Used for partial objects and after certain numbers.

Adessive: Indicates location on something.

Ablative: Indicates movement away from something.

Allative: Indicates movement towards something.

Essive: Indicates a temporary state or role.

Translative: Indicates change or transformation.

Instructive: Indicates the means or instrument.

Comitative: Indicates accompaniment.

Number

In grammar, number refers to the distinction between **singular**, **dual**, **plural** forms etc., which indicate the quantity of the subject or object in a sentence. These forms are primarily associated with nouns, pronouns, adjectives, and verbs, reflecting whether they refer to one, two, or more than two entities. E.g.:

In English: **singular** -- one cat, one dog, he; **dual&plural** -- two dogs, two cats, **his**.

In Ancient Greek: **singular** -- λύκο**ς**; **dual** -- λύκ**ω**; **plural** -- λύκο**ι**

In Sanskrit: **singular** -- पुरु**षः**; **dual** -- पुरु**षौ**; **plural** -- पुरु**षाः**

Number (cont.)

In many languages, grammatical number often interacts with other grammatical categories such as person, gender, and case. These interactions can result in complex rules for word forms and agreements.

E.g. in **English**, the verb "to be" changes its form based on person and number:

- First person **singular**: "I am"
- First person **plural**: "We are"
- Third person **singular**: "He/She/It is"
- Third person **plural**: "They are"

The influence of cultural and social background

Language and Culture

Language reflects the cultural background of its speakers.

For example, the Inuit language has many different words for snow, which reflects the significance of snow in their environment and daily life.

In English, there isn't a direct equivalent for the Chinese "羊", which broadly refers to sheep, goats, and other similar animals. English typically distinguishes between these animals using specific terms like "sheep" for sheep and "goat" for goats.

Cultural Exchange Shapes Language

Language and grammar changes due to cultural shock happen when people from different cultures interact and influence each other's ways of speaking. This can lead to new words, changes in how sentences are put together, and even mixing languages in the same conversation.

Borrowing Words: When new things, like foods or technologies, come from another culture, we often take the words for them. E.g. *Ketchup*, コンピューター, 走火入魔

New Grammar: Sometimes, people start using new ways of structuring sentences or phrases because of another language's influence. E.g. *long time no see*

Word Meaning Changes: Words can change meaning or gain new uses when cultures mix. For example, "*cool*" in English can mean both temperature and something trendy or popular; "*google*" is changed from noun to verb.

Languages Evolve Over Time

Languages evolve over time, influenced by historical events, social changes, and technological advancements. For example, the rise of the internet and social media has significantly impacted language, creating a wealth of new vocabulary and expressions. These changes reflect the new ways people communicate and interact in a digital world, leading to the adoption of terms like "tweet," "emoji," and "hashtag." As societies and technologies develop, languages continue to adapt, incorporating new concepts and forms of expression.

Conclusion

Pronunciation and Speech Sounds: Explored the production and classification of vowels and consonants, highlighting the diversity across languages.

Writing Systems: Discussed the variety of writing systems, including alphabetic, syllabic, and logographic systems.

Grammatical Features: Covered key grammatical concepts such as word order, tense and aspect, and grammatical mood across different languages.

Phonological Processes: Introduced common phonological processes like assimilation, elision, and epenthesis.

Cultural Influence on Language: Emphasized how language evolves due to cultural and social influences, including borrowing words and the creation of new grammar rules.